📝 Chapter 2: Mechanics of Futures Markets (Final Sections)

5. Over-the-Counter (OTC) Markets and Credit Risk

A. The Challenge: Credit Risk in OTC

- **Definition:** OTC markets are private agreements between two companies for derivative transactions **without using an exchange**.
- **The Problem:** Credit risk (or counterparty risk) is the main danger. If one party defaults on the contract, the other party loses money if the contract was favorable to them.
  - **Example:** Company A and Company B have an agreement. If the trade is currently worth **$5 million** to Company B, and Company A goes bankrupt, Company B takes a **$5 million loss**.
- **The Solution:** The OTC market has adopted centralized risk management techniques from the exchange-traded markets (futures).

B. Central Counterparties (CCPs)

- **Role:** CCPs act as **clearing houses for standardized OTC transactions**. They perform the same function as a futures exchange clearing house.
- **How it Works (Novation):** When a trade between A and B is accepted by a CCP, the CCP steps in and legally becomes the counterparty to both sides.
  - **Original Trade:** A agrees to buy an asset from B in one year.
  - **CCP View:** The CCP agrees to **buy the asset from B** and **sell the asset to A**.
- **Benefit:** The CCP takes on the credit risk of both A and B, effectively guaranteeing the performance of the trade.
- **Margin Requirements:** CCP members must provide **initial margin** and **daily variation margin** (just like futures) and contribute to a **guaranty fund**.
- **Regulatory Driver:** Post-2007 financial crisis, regulations were introduced to reduce systemic risk by **requiring** most standardized OTC derivatives (especially between major financial institutions) to be cleared through CCPs.
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C. Bilateral Clearing and Collateral

- **Definition:** Non-standard or exempt OTC transactions are still cleared **bilaterally** (directly between the two parties).
- **Master Agreement & CSA:** The parties sign an **ISDA Master Agreement**, often with a **Credit Support Annex (CSA)**, which requires the posting of collateral.
- **The Collateral Mechanism:** This is similar to margin, but it's called **collateral**. The key is that positions are valued *daily*:
  - **Example (Variation Margin):** Company A and Company B have a net exposure. If the mark-to-market value of all their trades increases by **$500,000** in favor of Company A today, the CSA requires Company B to post **$500,000 in collateral** to Company A.
  - This secures the exposure and drastically reduces the loss if the counterparty defaults.
- **Initial Margin (New Regulation):** Previously rare, new regulations require **initial margin** (in addition to variation margin collateral) for bilaterally cleared transactions between financial institutions. This initial margin must be **segregated** (held by a third party), providing an extra layer of safety.
- **LTCM Example (Danger of Leverage):** Hedge funds like LTCM used collateral agreements to achieve **extremely high leverage**. While collateral protected them from *credit risk*, the high leverage exposed them to **market risk**. When the Russian debt crisis caused massive market widening, LTCM was forced to post collateral on both long and short positions simultaneously, leading to a $4 billion loss.

D. Margin/Collateral Interest and Haircuts

- **Futures vs. OTC (Interest):**
  - **Futures Variation Margin:** Does **not** earn interest because it represents the daily settlement (daily cash flow).
  - **OTC Variation Margin/Collateral: Usually earns interest** because the underlying trade is **not settled daily**; the collateral is just security for a trade that continues to exist.
- **Haircut:** When securities (like bonds or stocks) are used for margin or collateral instead of cash, their value is reduced by a certain percentage—a **haircut**—to account for their potential price volatility.
  - **Example:** If you post a bond worth $\$10,000$ and the required haircut is 10%, the bond's collateral value is only **$9,000**.

6. Market Quotes

A. Understanding the Futures Quote Table

- Futures quotes are standardized by the exchange (e.g., CME Group).
- **Contract Specifications:**
  - **Gold:** 100 ounces, quoted in dollars per ounce.
  - **Crude Oil:** 1,000 barrels, quoted in dollars per barrel.

B. Analyzing Price Data (Using June 2013 Gold Example)

Column	Description	Example (June 2013 Gold)	Interpretation
Open, High, Low	Range of prices traded so far that day.	Open: $1,429.5. High: $1,444.9. Low: $1,419.7.	Shows the volatility and trading range of the contract during the session.
Prior Settlement	The price used for margin calculations from the previous day.	$1,434.3	This is the starting point for daily profit/loss calculation.
Last Trade	The most recent price at which the contract traded.	$1,425.3	
Change	The difference between the Last Trade and the Prior Settlement.	-9.0	**Example Impact:** A long trader loses $\$900$ ($9.0 loss $\times$ 100 oz contract size) on this day.
Trading Volume	Total number of contracts traded in the day.	147,943	
Open Interest	Total number of contracts outstanding (total long $\equiv$ total short).	*Not shown in this column, but is separate data.*	

C. Trading Volume vs. Open Interest

- **Key Distinction: Volume** is activity (contracts traded); **Open Interest** is inventory (contracts outstanding).
- **Day Trader Effect:** If there are many **day traders** (traders who enter and close out a position on the same day), the **Trading Volume can be greater than the Open Interest**.
  - **Example:** Open Interest is 10,000 contracts. A day trader buys 10 contracts and sells them before the close. Trading Volume increases by 10 (buy) + 10 (sell) = 20, but Open Interest remains 10,000.

D. Patterns of Futures Prices

- **Normal Market (Contango):** Futures prices are an **increasing function of maturity**.
  - **Example:** Wheat futures are an increasing function of maturity (July 2013: 710.00 cents; March 2014: 752.50 cents). The market expects a slow rise in price over time.
- **Inverted Market (Backwardation):** Futures prices **decline with maturity**.
  - **Example:** Crude oil prices often show this, indicating high current demand but expecting the price to fall as time passes and supply eases.

7. Delivery and Cash Settlement

A. Delivery Procedure (The Physical Exchange)

- **Rarity:** Very few contracts lead to delivery; most are **closed out** before the delivery period.
- **Initiation:** The party with the **short position** (the seller) decides when to deliver within the defined delivery period.
- **Notice and Assignment:** The short seller issues a **notice of intention to deliver**. The exchange assigns this notice to the party with the **oldest outstanding long position**.
  - **Note:** The person who takes delivery is rarely the person who originally entered the contract with the seller.

- **Payment and Price:** The buyer (long position) makes immediate payment at the **most recent settlement price** (adjusted for grade or location).
    - **Commodity:** Delivery means accepting a **warehouse receipt** (and the buyer assumes warehousing costs).
    - **Financial Futures:** Delivery is usually done via **wire transfer**.

B. Critical Delivery Dates

To manage the risk of having to deliver or take delivery, three dates are critical:

1. **First Notice Day:** The first day the short position can submit a delivery notice.
2. **Last Notice Day:** The last day the short position can submit a delivery notice.
3. **Last Trading Day:** The last day the contract can be traded. (This is generally a few days *before* the First Notice Day).

- **Investor Tip:** If you have a **long position** and do not want to take physical delivery, you **must close out your position before the First Notice Day**.

C. Cash Settlement

- **When Used:** Used when physical delivery is **impractical or impossible**.
    - **Example:** Delivering the underlying asset for the **S&P 500 futures** would mean delivering a proportional portfolio of **500 individual stocks**, which is too complex.
- **Mechanism:** On a predetermined day (e.g., the third Friday of the delivery month for S&P 500 futures), all outstanding contracts are simply **closed in cash**.
- **Final Price:** The final settlement price is set equal to the **spot price** of the underlying asset at the open or close of trading on that predetermined day. The resulting gain or loss is paid out through the margin accounts.